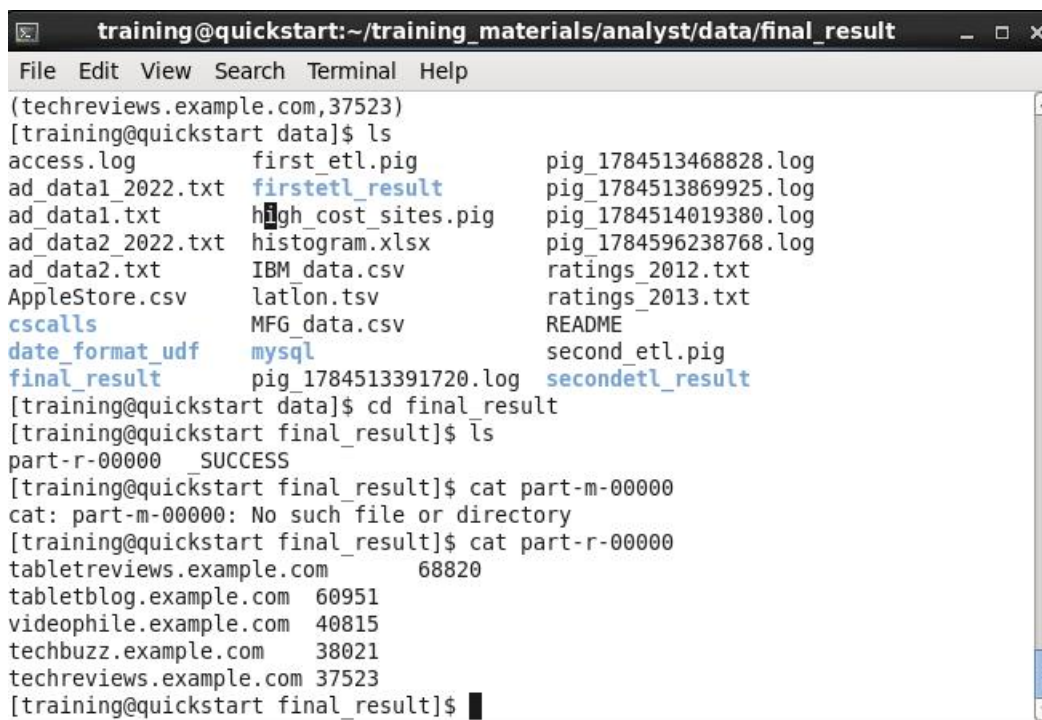


Lab Task 5
Analysing Ad campaign data with Pig
COS20028 – Big data Architecture and Applications
Semester 2

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Screenshot of the First Page Result Content



```
training@quickstart:~/training_materials/analyst/data/final_result
File Edit View Search Terminal Help
(techreviews.example.com,37523)
[training@quickstart data]$ ls
access.log          first_etl.pig          pig_1784513468828.log
ad_data1_2022.txt   firstetl_result        pig_1784513869925.log
ad_data1.txt        high_cost_sites.pig    pig_1784514019380.log
ad_data2_2022.txt   histogram.xlsx         pig_1784596238768.log
ad_data2.txt        IBM_data.csv           ratings_2012.txt
AppleStore.csv     latlon.tsv             ratings_2013.txt
cscalls            MFG_data.csv           README
date_format_udf     mysql                  second_etl.pig
final_result        pig_1784513391720.log  secondetl_result
[training@quickstart data]$ cd final_result
[training@quickstart final_result]$ ls
part-r-000000_SUCCESS
[training@quickstart final_result]$ cat part-m-000000
cat: part-m-000000: No such file or directory
[training@quickstart final_result]$ cat part-r-000000
tabletreviews.example.com      68820
tabletblog.example.com        60951
videophile.example.com        40815
techbuzz.example.com          38021
techreviews.example.com       37523
[training@quickstart final_result]$
```

The Code for Processing the First AD:
(The “first_etl.pig” content)

```

GNU nano 2.0.9                                File: first etl.pig
raw_data1 = LOAD '/home/training/training_materials/analyst/data/ad_data1.txt' USING PigStorage('\t')
AS (keyword:chararray, campaign_id:chararray, date:chararray, time:chararray,
    display_site:chararray, was_clicked:int, cpc:int, country:chararray, placement:chararray);

filtered_data1 = FILTER raw_data1 BY (was_clicked == 1) AND (country == 'France');
projected_data1 = FOREACH filtered_data1 GENERATE display_site, cpc;

RMF /home/training/training_materials/analyst/data/firstetl_result;
STORE projected_data1 INTO '/home/training/training_materials/analyst/data/firstetl_result' USING PigStorage('\t');

```

The Code for Processing the Second AD: (The “second_etl.pig” content)

```

raw_data2 = LOAD '/home/training/training_materials/analyst/data/ad_data2.txt' USING PigStorage('\t')
AS (campaign_id:chararray, date:chararray, time:chararray, display_site:chararray,
    placement:chararray, was_clicked:int, cpc:int, keyword:chararray);

filtered_data2 = FILTER raw_data2 BY was_clicked == 1;
projected_data2 = FOREACH filtered_data2 GENERATE display_site, cpc;

RMF /home/training/training_materials/analyst/data/secondetl_result;
STORE projected_data2 INTO '/home/training/training_materials/analyst/data/secondetl_result' USING PigStorage('\t');

```

The Code for Merging Multiple Sources and Finding Answers: (The “high_cost_sites.pig” content)

```

GNU nano 2.0.9                                File: high cost sites.pig
data1 = LOAD '/home/training/training_materials/analyst/data/firstetl_result' USING PigStorage('\t') AS (display_site:chararray, cpc:int);
data2 = LOAD '/home/training/training_materials/analyst/data/secondetl_result' USING PigStorage('\t') AS (display_site:chararray, cpc:int);

all_data = UNION data1, data2;
grouped_by_site = GROUP all_data BY display_site;
site_costs = FOREACH grouped_by_site GENERATE group AS display_site, SUM(all_data.cpc) AS total_cost;
sorted_costs = ORDER site_costs BY total_cost DESC;
top5_websites = LIMIT sorted_costs 5;

RMF /home/training/training_materials/analyst/data/final_result;
STORE top5_websites INTO '/home/training/training_materials/analyst/data/final_result' USING PigStorage('\t');
DUMP top5_websites;

```